



INDIAN INSTITUTE OF INFORMATION TECHNOLOGY, SRI CITY

Machine Learning Project: UG 3_CSE M2026

Deadline: 31st October, 2026

1. Use a standard dataset and compare the following classifiers. **[10 Marks]**
 - a. K-Nearest Neighbour Classifier with cross-validation
 - b. Decision Tree
 - c. Support Vector Machine with RBF Kernel

Implementation Details:

- i) Use Train-test split as 80% Train and 20% test.
- ii) Generate the confusion matrix.
- iii) Compare accuracy, precision, recall, F1-score and ROC-AUC.
- iv) Discuss the effect of overfitting and model complexity.
- v) State which classifier is best among the given classifiers based on ROC curve analysis.

2. Use a standard dataset and compare the following regression models **[10 Marks]**
 - a. Linear regression with gradient descent optimization
 - b. Polynomial Regression
 - c. Ridge and Lasso Regressions

Implementation Details:

- i) Calculate MSE, RMSE and R^2 for each regression model and compare them.
- ii) Show overfitting and underfitting cases.
- iii) Use cross-validation to select the regularization parameter.

3. a) Use a standard dataset and implement K-Means Clustering from scratch. **[5 Marks]**

Implementation Details:

- i) Experiment with different numbers of clusters: $K=2,3,4,5,6$.
- ii) For each value of K , calculate the Within-Cluster Sum of Squares (WCSS).
- iii) Plot the Elbow curve.

- b) Implement PCA using the covariance matrix and eigenvalue decomposition. **[5 Marks]**

Implementation Details:

- i) Calculate the eigenvalues and eigenvectors.
- ii) Determine the explained variance ratio.
- iii) Plot cumulative explained variance.
- iv) Reduce the original feature space to:
 - 2 principal components
 - 3 principal components
- v) Visualize the data using the first two principal components.

Submission: Write a report on the above assignments and upload a pdf file. Submit the codes in zip.